

Author Contributions Checklist Form

This form documents the artifacts associated with the article (i.e., the data and code supporting the computational findings) and describes how to reproduce the findings.

Part 1: Data

☐ This paper **does not** involve analysis of external data (i.e., no data are used or the only data are generated by the authors via simulation in their code).

☒ I certify that the author(s) of the manuscript have legitimate access to and permission to use the data used in this manuscript.

Abstract

Our data analysis involves a multi-site evaluation of the effectiveness of GLP-1 RAs in preventing mental health conditions, based on the EHR data at the University of Pennsylvania Health System. Due to institutional privacy policies and data use agreements, these datasets cannot be shared publicly. A synthetic dataset that mimics the structure and distributional characteristics of the data is provided.

Availability

- ☐ Data **are** publicly available
- ☒ Data **cannot be made** publicly available

If the data are publicly available, see the *Publicly available data* section. Otherwise, see the *Non-publicly available data* section, below.

Publicly available data

- ☐ Data are available online at:
- ☐ Data are available as part of the paper's supplementary material.
- ☐ Data are publicly available by request, following the process described here:

- ☐ Data are or will be made available through some other mechanism, described here:

Non-publicly available data

Discussion of lack of publicly available data:

Due to institutional privacy policies and data use agreements, the patient-level data used in the manuscript cannot be shared publicly. To facilitate transparency and reproducibility, a synthetic dataset that mimics the structure and distributional characteristics of the data is provided. This synthetic dataset is sufficient to illustrate the proposed approach. While results obtained from the synthetic data are not numerically identical to those from the real data, they adequately demonstrate the effectiveness and advantages of the proposed method.

Description

File format(s)

- ☐ CSV or other plain text:
- ☐ Software-specific binary format (.Rda, Python pickle, etc.):
- ☐ Standardized binary format (e.g., netCDF, HDF5, etc.):
- ☒ Other (described here):

.rdata file

Data dictionary

- ☐ Provided by the authors in the following file(s):
- ☒ Data file(s) is (are) self-describing (e.g., netCDF files)
- ☐ Available at the following URL:

Additional information (optional)

Part 2: Code

Abstract

We provide the R code used to conduct the simulation study in Section 6 of the manuscript. The results reported in Table 1 can be reproduced by executing `main_simu.r`.

We also provide the R code for the use case analyzed in Section 7 of the manuscript, which can be run based on the provided synthetic dataset. While results obtained from the synthetic data are not numerically identical to those from the real data, they adequately demonstrate the effectiveness and advantages of the proposed method.

Description

Code format(s)

☒ Script files

☒ R ☐ Python ☐ Matlab

☐ Other:

☐ Package

☐ R ☐ Python ☐ MATLAB toolbox

☐ Other:

☐ Reproducible report

☐ R Markdown ☐ Jupyter notebook

☐ Other:

☐ Shell script

☐ Other (described here):

Supporting software requirements

Version of primary software used

R 4.3.1

Libraries and dependencies used by the code

parallel; xtable; mvtnorm; MASS; survival; pracma; ggplot2

Supporting system/hardware requirements (optional)

Parallelization used

- ☐ No parallel code used
- ☒ Multi-core parallelization on a single machine/node
Number of cores used: 2
- ☐ Multi-machine/multi-node parallelization
Number of nodes and cores used:

License

- ☒ MIT License (default)
- ☐ BSD
- ☐ GPL v3.0
- ☐ Creative Commons
- ☐ Other (described here):

Additional information (optional)

Part 3: Reproducibility workflow

Scope

The provided workflow reproduces:

- ☐ Any numbers provided in text in the paper
- ☒ The computational method(s) presented in the paper (i.e., code is provided that implements the method(s))
- ☐ All tables and figures in the paper
- ☒ Selected tables and figures in the paper, as explained and justified here:

We provide codes to reproduce our main simulation results in Table 1 of the manuscript. Simulation results in the supplementary materials can be readily reproduced based on the provided code and only requires some minor adjustment.

We also provide the R code for the use case analyzed in Section 7 of the manuscript, which can be run based on the provided synthetic dataset. While results obtained from the synthetic data are not numerically identical to those from the real data, they adequately demonstrate the effectiveness and advantages of the proposed method.

Workflow details

Location

The workflow is available:

- ☒ As part of the paper's supplementary material
- ☒ In this Git repository: <https://github.com/submission-simu/simulation-study>
- ☐ Other:

Format(s)

- ☐ Single master code file
- ☐ Wrapper (shell) script(s)
- ☐ Self-contained R Markdown file, Jupyter notebook, or other literate programming approach
- ☒ Text file (e.g., a readme-style file) that documents workflow
- ☐ Makefile
- ☐ Other (more detail in 'Instructions' below)

Instructions

All instructions are provided in the README file.

Expected run-time

Approximate time needed to reproduce the analyses on a standard desktop machine:

- ☐ <1 minute
- ☐ 1-10 minutes
- ☒ 10-60 minutes
- ☐ 1-8 hours
- ☐ >8 hours
- ☐ Not feasible to run on a desktop machine, as described here:

Additional documentation (optional)

Notes (optional)